

KUGLUKTUK, NU, Canada

WMO: 713320

Lat: 67.817N

Lon: 115.135W

Elev: 74

StdP: 14.66

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.4	-36.3	-44.3	0.4	-35.7	-41.3	0.5	-33.6	32.3	-0.2	29.2	-1.9	6.2	230	0.968

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	73.2	59.2	67.7	57.0	62.9	54.8	60.4	70.5	57.9	66.5	55.3	62.2	10.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.9	66.7	64.0	53.3	60.6	61.8	50.9	55.5	59.0	26.8	70.7	25.1	66.9	23.4	62.3	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	23.2	20.5	-43.4	82.1	3.6	3.9	-45.9	85.0	-48.0	87.3	-50.0	89.5	-52.6	92.3		
(5)			-40.5	64.7	3.0	2.5	-42.6	66.5	-44.3	67.9	-46.0	69.3	-48.1	71.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	-15.7	-16.1	-12.4	2.8	23.6	42.2	52.0	49.1	38.8	21.7	1.9	-11.0		
	DBStd	27.16	12.08	11.99	12.14	12.25	10.39	8.00	7.00	7.05	7.18	9.20	10.75	11.76		
	HDD50	13037	2035	1851	1936	1417	820	262	58	102	343	878	1444	1891		
	HDD65	18292	2500	2271	2401	1867	1285	685	408	497	785	1343	1894	2356		
	CDD50	227	0	0	0	0	0	28	119	73	7	0	0	0		
	CDD65	8	0	0	0	0	0	1	4	3	0	0	0	0		
	CDH74	121	0	0	0	0	0	24	58	39	0	0	0	0		
	CDH80	19	0	0	0	0	0	5	9	5	0	0	0	0		
	WSAvg	9.5	11.1	10.4	9.5	8.3	8.4	8.2	8.5	9.0	10.1	10.0	10.6	10.2		
	PrecAvg	9.3	0.4	0.3	0.4	0.5	0.5	0.6	1.4	1.6	1.3	0.9	0.5	0.4		
(15) Precipitation	PrecMax	14.0	1.3	1.0	0.6	2.4	1.8	1.8	3.0	5.2	4.1	2.4	1.1	1.1		
	PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.3	0.1	0.0	0.0	0.0		
	PrecStd	2.8	0.3	0.2	0.2	0.5	0.4	0.5	0.7	1.1	0.9	0.6	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	17.4	25.0	37.2	53.5	77.5	80.4	79.5	64.7	44.4	30.6	21.5		
		MCWB	21.5	16.2	23.5	34.4	45.5	58.4	61.7	61.7	54.6	41.3	29.4	20.0		
	2%	DB	13.3	11.7	18.1	32.2	45.7	67.8	73.9	72.0	58.0	37.7	25.5	16.3		
		MCWB	12.3	10.8	16.8	30.5	40.6	54.3	59.8	59.4	50.7	35.8	24.2	15.2		
	5%	DB	6.9	6.5	11.9	26.4	40.3	61.6	68.8	65.7	53.2	35.4	21.5	11.4		
		MCWB	6.0	5.5	11.0	24.8	36.9	52.4	58.1	56.9	48.0	34.0	20.5	10.5		
	10%	DB	1.8	2.3	5.7	21.0	36.8	55.4	63.9	60.3	49.1	33.2	17.1	6.4		
		MCWB	1.0	1.6	4.9	19.8	34.5	48.7	55.8	53.9	45.5	32.0	16.0	5.6		
	0.4%	WB	21.6	16.5	23.5	34.5	45.8	59.1	64.2	63.3	55.9	41.2	29.3	20.3		
		MCDB	22.6	17.5	25.0	37.1	53.1	73.3	76.2	74.7	62.4	44.5	30.4	21.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	12.5	11.0	16.9	30.2	40.7	55.9	61.2	60.2	51.3	36.3	24.3	15.3		
		MCDB	13.4	11.8	18.1	32.3	45.3	66.5	71.0	70.4	57.2	37.5	25.3	16.2		
	5%	WB	6.3	5.8	11.1	25.0	37.1	52.4	58.9	57.2	48.3	34.1	20.5	10.6		
		MCDB	7.2	6.6	12.0	26.5	39.9	61.0	67.3	64.6	52.2	35.3	21.4	11.4		
	10%	WB	1.3	1.8	5.0	19.8	34.6	48.7	56.4	54.3	45.8	32.0	16.1	5.8		
		MCDB	1.9	2.5	5.7	21.0	36.7	55.4	63.5	60.2	49.0	33.0	17.0	6.5		
	5% DB	MDBR	12.8	13.0	13.7	14.8	12.4	14.1	14.9	13.1	10.1	9.5	12.1	12.7		
		MCDBR	16.9	17.3	17.9	17.2	17.3	26.7	24.0	23.2	18.8	8.7	14.0	16.7		
		MCWBR	16.6	17.0	17.4	16.1	13.1	16.1	12.8	12.9	12.2	7.8	13.7	16.3		
		MCWBR	16.9	17.1	17.7	16.5	16.8	25.4	21.6	21.3	17.0	8.7	13.9	16.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.6	16.8	17.2	15.5	12.9	16.1	12.8	12.4	11.4	7.8	13.6	16.4		
(40) Clear-Sky Solar Irradiance	taub		0.161	0.228	0.253	0.297	0.335	0.347	0.364	0.340	0.296	0.247	0.165	N/A		
	taud		1.894	2.060	2.207	2.143	2.177	2.339	2.368	2.467	2.529	2.210	1.914	N/A		
	Ebn at Noon		49	186	254	269	270	271	261	256	244	177	41	0		
	Edh at Noon		3	15	24	35	38	33	32	26	18	13	3	0		
(44) All-Sky Solar Radiation	RadAvg		14	180	689	1454	1971	1861	1652	1111	560	173	33	0		
	RadStd		1	13	43	67	132	116	121	128	67	27	2	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page